

1. TemperatureUnitConverter

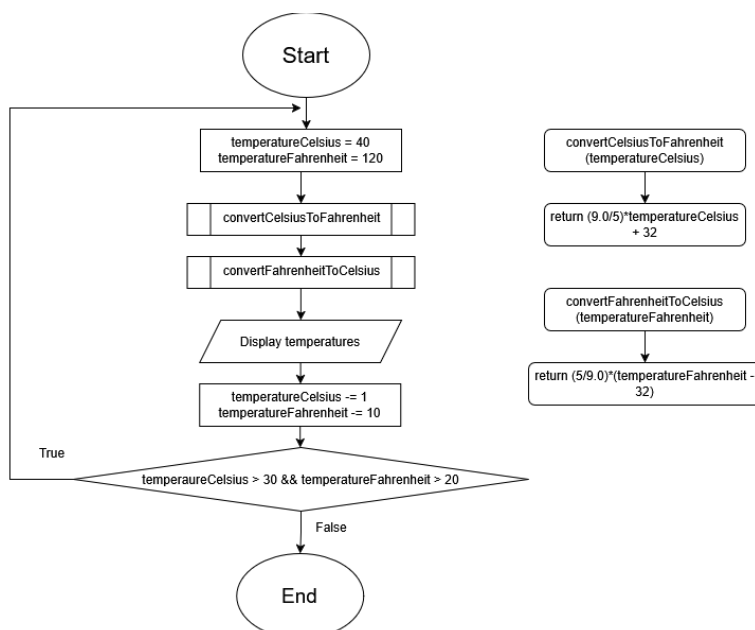
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Temperature in Fahrenheit 2. Temperature in Celsius	1. Temperature in Celsius 2. Temperature in Fahrenheit
Processing Required	Solution Alternative
Set temperatureFahrenheit, temperatureCelsius temperatureFahrenheit = $(9.0/5)*\text{temperatureCelsius} + 32$ temperatureCelsius = $(5/9.0)*(\text{temperatureFahrenheit} - 32)$ Display table	1. Define temperature in Fahrenheit and Celsius as constants. 2. Define temperature in Fahrenheit and Celsius as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Temperature in Fahrenheit 2. Temperature in Celsius	1. Set temperatureFahrenheit from 120 to 30 2. Set temperatureCelsius from 40 to 31 3. Method for temperatureFahrenheit = $(9.0/5)*\text{temperatureCelsius} + 32$ 4. Method for temperatureCelsius = $(5/9.0)*(\text{temperatureFahrenheit} - 32)$ 5. Use method to convert temperature unit 6. Display in a table 7. End	1. Main 2. convertCelsiusToFahrenheit 3. convertFahrenheitToCelsius	1. Temperature in Celsius 2. Temperature in Fahrenheit

c) Flowchart



d) Pseudocode

Main Module

```
double temperatureCelsius = 40;
double temperatureFahrenheit = 120;
do{
    Process convertCelsiusToFahrenheit
    Process convertFahrenheitToCelsius
    Display temperatures
    temperatureCelsius -= 1;
    temperatureFahrenheit -= 10;
}while(temperatureCelsius > 30 && temperatureFahrenheit > 20)
```

End

convertCelsiusToFahrenheit (temperatureCelsius) Module

```
return (9.0/5)*temperatureCelsius + 32
```

Exit

convertFahrenheitToCelsius (temperatureFahrenheit) Module

```
return (5/9.0)*(temperatureFahrenheit - 32)
```

Exit

e) Java code

```
// NGO YUK HANG 228991 LAB 7 Q1
public class TemperatureUnitConverter{
    public static void main(String[] args){
        System.out.printf("%-10s %-10s | %-10s %-10s\n", "Celsius", "Fahrenheit", "Fahrenheit", "Celsius");
        double temperatureCelsius = 40.00;
        double temperatureFahrenheit = 120.00;
        do{
            System.out.printf("%-10.2f %-10.2f | %-10.2f %-10.2f\n", temperatureCelsius, convertCelsiusToFahrenheit(temperatureCelsius), temperatureFahrenheit, convertFahrenheitToCelsius(temperatureFahrenheit));
            temperatureCelsius -= 1;
            temperatureFahrenheit -= 10;
        } while(temperatureCelsius > 30 && temperatureFahrenheit > 20);
    }

    public static double convertCelsiusToFahrenheit(double temperatureCelsius){
        return (9.0/5) * temperatureCelsius + 32;
    }

    public static double convertFahrenheitToCelsius(double temperatureFahrenheit){
        return (temperatureFahrenheit - 32) * (5/9.0);
    }
}
```

2. Lab7Q2

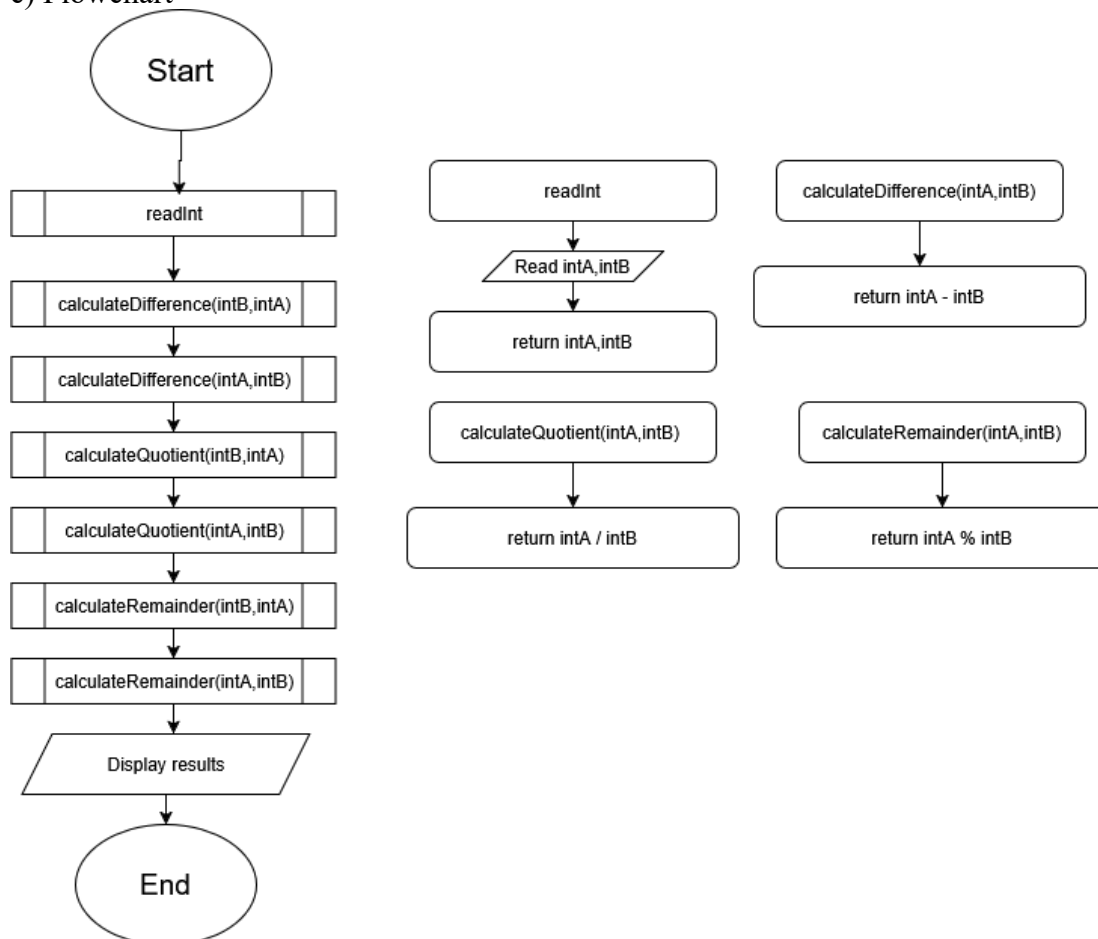
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. intA 2. intB	1. Differences 2. Quotients 3. Remainders
Processing Required	Solution Alternative
1. Read intA, intB 2. Calculate intB - intA, intA - intB, intB/intA, intA/intB, intB%intA, intA%intB 3. Display result	1. Define intA, intB as constants. 2. Define intA, intB as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. intA 2. intB	1. Read intA, intB 2. Method to calculate intB - intA, intA - intB, intB/intA, intA/intB, intB%intA, intA%intB 3. Display result 4. End	1. readInt 2. calculateDifference 3. calculateQuotient 4. calculateRemainder	1. Differences 2. Quotients 3. Remainders. Total ticket price

c) Flowchart



d) Pseudocode

Main Module

```
Process readInt
Process calculateDifference(intB,intA)
Process calculateDifference(intA,intB)
Process calculateQuotient(intB,intA)
Process calculateQuotient(intA,intB)
Process calculateRemainder(intB,intA)
Process calculateRemainder(intA,intB)
Display result
```

End

readInt Module

```
Read intA,intB
return intA,intB
```

Exit

calculateDifference(intA,intB) Module

```
return intA - intB
```

Exit

calculateQuotient(intA,intB) Module

```
return intA / intB
```

Exit

calculateRemainder(intA,intB) Module

```
return intA % intB
```

Exit

e) Java code

```
import java.util.Scanner;
import java.util.InputMismatchException;

public class Lab7Q2{

    public static Scanner scanner = new Scanner(System.in);
    public static int result;

    public static void main(String[] args){
        System.out.println("Please enter 2 integers to calculate their differences, quotients and remainders");

        int intA = readInt("First integer : ");
        int intB = readInt("Second integer : ");

        System.out.println();

        calculateDifference(intB,intA);
        System.out.println(intB + "-" + intA + "=" + result);

        calculateDifference(intA,intB);
        System.out.println(intA + "-" + intB + "=" + result);

        calculateQuotient(intB,intA);
        System.out.println(intB + "/" + intA + "=" + result);

        calculateQuotient(intA,intB);
        System.out.println(intA + "/" + intB + "=" + result);

        calculateRemainder(intB,intA);
        System.out.println(intB + "%" + intA + "=" + result);

        calculateRemainder(intA,intB);
        System.out.println(intA + "%" + intB + "=" + result);
    }

    public static int readInt(String message){
        int input;
        while(true){
            System.out.println(message);
            try{
                input = scanner.nextInt();
            } catch (InputMismatchException e){
                System.out.println("Please enter an integer.");
                scanner.nextLine();
                continue;
            }
            break;
        }
        return input;
    }

    public static void calculateDifference(int first, int second){
        result = first - second;
    }

    public static void calculateQuotient(int first, int second){
        result = first / second;
    }

    public static void calculateRemainder(int first, int second){
        result = first % second;
    }
}
```

3. GradeCalculatorWithMethods (continue from Lab4Q3)

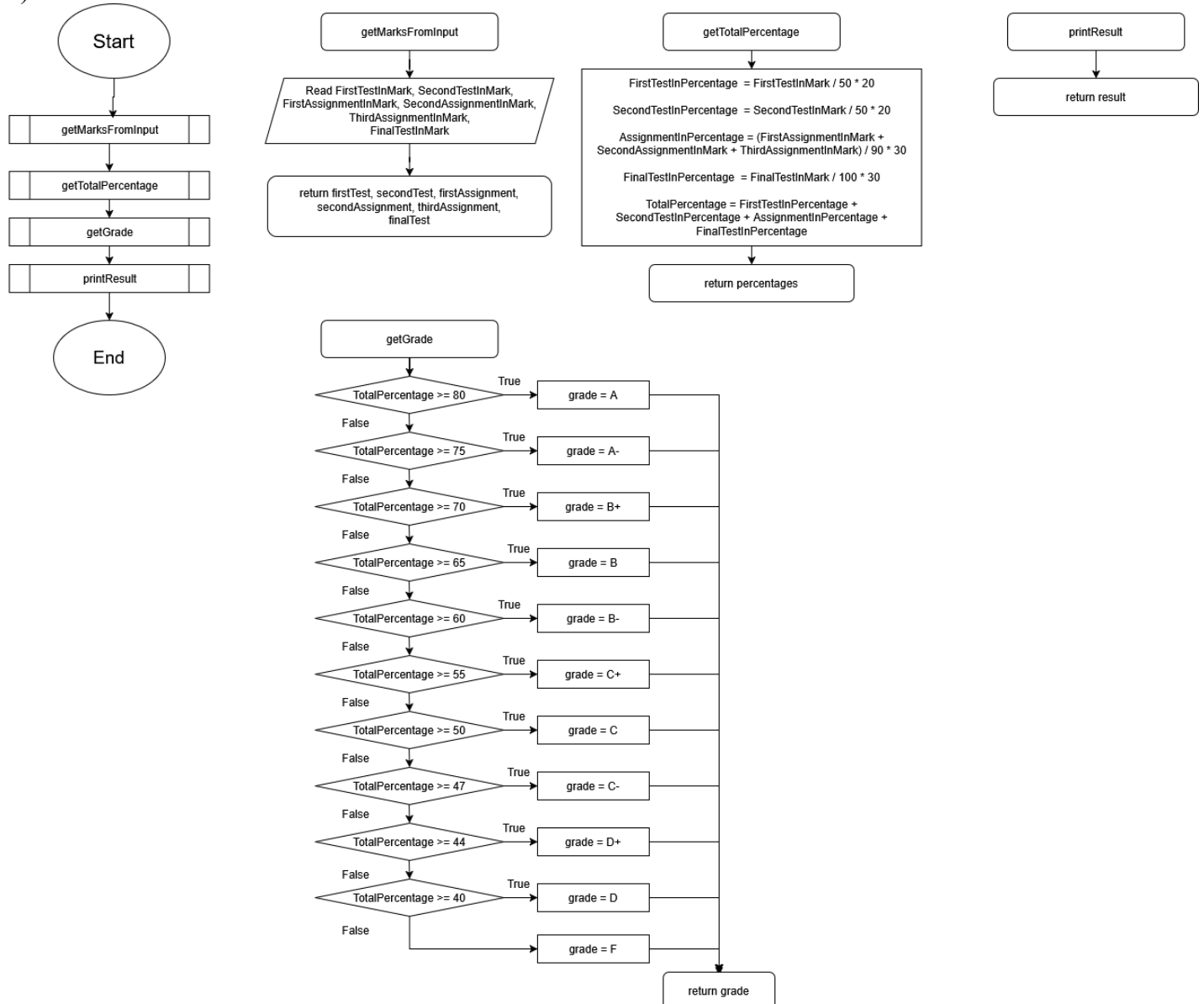
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Mark for first test 2. Mark for second test 3. Mark for assignments 4. Mark for final exam	1. All marks in raw mark 2. Total mark in percentage 3. Grade
Processing Required	Solution Alternative
$\text{FirstTestInPercentage} = \text{FirstTestInMark} / 50 * 20$ $\text{SecondTestInPercentage} = \text{SecondTestInMark} / 50 * 20$ $\text{AssignmentInPercentage} = (\text{FirstAssignmentInMark} + \text{SecondAssignmentInMark} + \text{ThirdAssignmentInMark}) / 90 * 30$ $\text{FinalTestInPercentage} = \text{FinalTestInMark} / 100 * 30$ $\text{TotalPercentage} = \text{FirstTestInPercentage} + \text{SecondTestInPercentage} + \text{AssignmentInPercentage} + \text{FinalTestInPercentage}$ if(TotalPercentage in some ranges) Grade = A/A-/B+/B/B-/C+/C/C-/D+/D/F	1. Define marks for each tests as constants. 2. Define marks for each tests as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. FirstTest 2. SecondTest 3. First Assignment 4. Second Assignment 5. Third Assignment 6. FinalTest	1. Method to read marks 2. Method to calculate percentage of each test and totalPercentage 3. Method to calculate grade 4. Method to display all results 5. End	1. getMarksFromInput 2. getTotalPercentage 3. getGrade 4. printResult	1. Raw marks for each tests 2. TotalPercentage 3. Grade

c) Flowchart



d) Pseudocode

Main Module

```
    Process getMarksFromInput
    Process getTotalPercentage
    Process getGrade
    Process printResult
```

End

getMarksFromInput Module

```
    Read FirstTestInMark, SecondTestInMark, FirstAssignmentInMark, SecondAssignmentInMark,
        ThirdAssignmentInMark, FinalTestInMark
    Return FirstTestInMark, SecondTestInMark, FirstAssignmentInMark,
        SecondAssignmentInMark, ThirdAssignmentInMark, FinalTestInMark
```

Exit

getTotalPercentage Module

```
    FirstTestInPercentage = FirstTestInMark / 50 * 20
    SecondTestInPercentage = SecondTestInMark / 50 * 20
    AssignmentInPercentage = (FirstAssignmentInMark + SecondAssignmentInMark +
        ThirdAssignmentInMark) / 90 * 30
    FinalTestInPercentage = FinalTestInMark / 100 * 30
    TotalPercentage = FirstTestInPercentage + SecondTestInPercentage + AssignmentInPercentage
        + FinalTestInPercentage
    Return percentages
```

Exit

getGrade Module

```
    if (totalPercentage >= 80)
        grade = "A";
    else if (totalPercentage >= 75)
        grade = "A-";
    else if (totalPercentage >= 70)
        grade = "B+";
    else if (totalPercentage >= 65)
        grade = "B";
    else if (totalPercentage >= 60)
        grade = "B-";
    else if (totalPercentage >= 55)
        grade = "C+";
    else if (totalPercentage >= 50)
        grade = "C";
    else if (totalPercentage >= 47)
        grade = "C-";
    else if (totalPercentage >= 44)
        grade = "D+";
    else if (totalPercentage >= 40)
        grade = "D";
    else
```


grade = "F";

Return grade

Exit

printResult Module

Display results

Exit

e) Java code

```
import java.util.Scanner;
import java.util.InputMismatchException;
//NGO YUK HANG 228991 LAB 7 Q3
public class GradeCalculatorWithMethods{

    public static Scanner scanner = new Scanner(System.in);
    public static int input;

    public static String[] markMessages = {
        "First test (Full mark is 50) : ",
        "Second test (Full mark is 50) : ",
        "First assignment (Full mark is 30) : ",
        "Second assignment (Full mark is 30) : ",
        "Third assignment (Full mark is 30) : ",
        "Final test (Full mark is 100) : "
    };

    public static final int[] fullMarks = {50,50,30,30,30,100};
    public static final int[] markToPercentage = {20,20,30,30,30,30};
    public static int[] marks = new int[6];
    public static int[] percentages = new int[4];
    public static int totalPercentage;
    public static String grade;

    public static void main(String[] args){
        // Print welcome message
        System.out.println("Welcome to grade calculator!");

        getMarksFromInput();

        getTotalPercentage();

        getGrade();

        printResult();
    }

    public static void getMarksFromInput(){
        // Get raw marks from user
        System.out.println("Please enter the marks for each tests below.");
        boolean success;
        for (int i = 0; i < 6; i++){
            do{
                System.out.println(markMessages[i]);
                success = getAndCheckMark(fullMarks[i]);
                if (success){
                    marks[i] = input;
                }
            } while (!success);
        }
    }

    public static boolean getAndCheckMark(int fullMark){
        try {
            input = scanner.nextInt();
        } catch (InputMismatchException e){
            System.out.println("Please enter an integer.");
            scanner.nextLine();
            return false;
        }
        if (input > fullMark || input < 0){
            System.out.println("You enter a wrong number. Please try again.");
            return false;
        }
        return true;
    }
}
```

```

public static void getTotalPercentage(){
    // Convert marks to percentage form
    // First test
    percentages[0] = marks[0] * markToPercentage[0] / fullMarks[0];
    // Second test
    percentages[1] = marks[1] * markToPercentage[1] / fullMarks[1];
    // Assignments
    percentages[2] = (marks[2] + marks[3] + marks[4]) * markToPercentage[2] / (fullMarks[2] + fullMarks[3] + fullMarks[4]);
    // Final test
    percentages[3] = marks[5] * markToPercentage[3] / fullMarks[5];

    // Calculate total percentage
    totalPercentage = percentages[0] + percentages[1] + percentages[2] + percentages[3];
}

public static void getGrade(){
    // Find grade
    if (totalPercentage ≥ 80)
        grade = "A";
    else if (totalPercentage ≥ 75)
        grade = "A-";
    else if (totalPercentage ≥ 70)
        grade = "B+";
    else if (totalPercentage ≥ 65)
        grade = "B";
    else if (totalPercentage ≥ 60)
        grade = "B-";
    else if (totalPercentage ≥ 55)
        grade = "C+";
    else if (totalPercentage ≥ 50)
        grade = "C";
    else if (totalPercentage ≥ 47)
        grade = "C-";
    else if (totalPercentage ≥ 44)
        grade = "D+";
    else if (totalPercentage ≥ 40)
        grade = "D";
    else
        grade = "F";
}

public static void printResult(){
    // Print outputs
    System.out.println("");
    System.out.println("Result : ");
    System.out.println("");
    for (int i = 0; i < 6; i++){
        System.out.println(markMessages[i] + " " + marks[i]);
    }
    System.out.println("");
    System.out.println("Total percentage : " + totalPercentage + "%");
    System.out.println("Grade : " + grade);
}
}

```

4. AverageCalculator

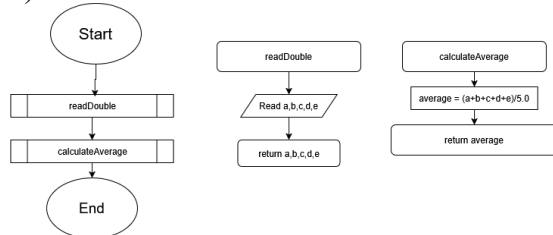
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. a 2. b 3. c 4. d 5. e	1. Average
Processing Required	Solution Alternative
Read a,b,c,d,e $\text{average} = (a+b+c+d+e)/5.0$ Display average	1. Define a,b,c,d,e as constants. 2. Define a,b,c,d,e as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. a 2. b 3. c 4. d 5. e	1. Method to read a,b,c,d,e 2. Method to calculate average 3. Display average 4. End	1. readDouble 2. calculateAverage	1. Average

c) Flowchart



d) Pseudocode

Main Module

Process readDouble
 Process calculateAverage

End

readDouble Module

Read a,b,c,d,e
 Return a,b,c,d,e

Exit

calculateAverage Module

$\text{average} = (a+b+c+d+e)/5.0$
 return average

Exit

e) Java code

```
import java.util.Scanner;
import java.util.InputMismatchException;
// NGO YUK HANG 228991 LAB 7 Q4
public class AverageCalculator{
    public static void main(String[] args){
        System.out.println("Please enter 5 numbers to calculate their average.");

        Scanner scanner = new Scanner(System.in);
        double a = readDouble("Enter first number : ",scanner);
        double b = readDouble("Enter second number : ",scanner);
        double c = readDouble("Enter third number : ",scanner);
        double d = readDouble("Enter fourth number : ",scanner);
        double e = readDouble("Enter fifth number : ",scanner);

        double average = calculateAverage(a,b,c,d,e);

        System.out.println("\nAverage : " + average);
    }

    public static double readDouble(String message, Scanner scanner){
        double input;
        while(true){
            System.out.println(message);
            try{
                input = scanner.nextDouble();
            } catch(InputMismatchException e){
                System.out.println("Please enter a double.");
                scanner.nextLine();
                continue;
            }
            break;
        }
        return input;
    }

    public static double calculateAverage(double a,double b,double c,double d,double e){
        return (a+b+c+d+e)/5.0;
    }
}
```

5. MoneyConverter

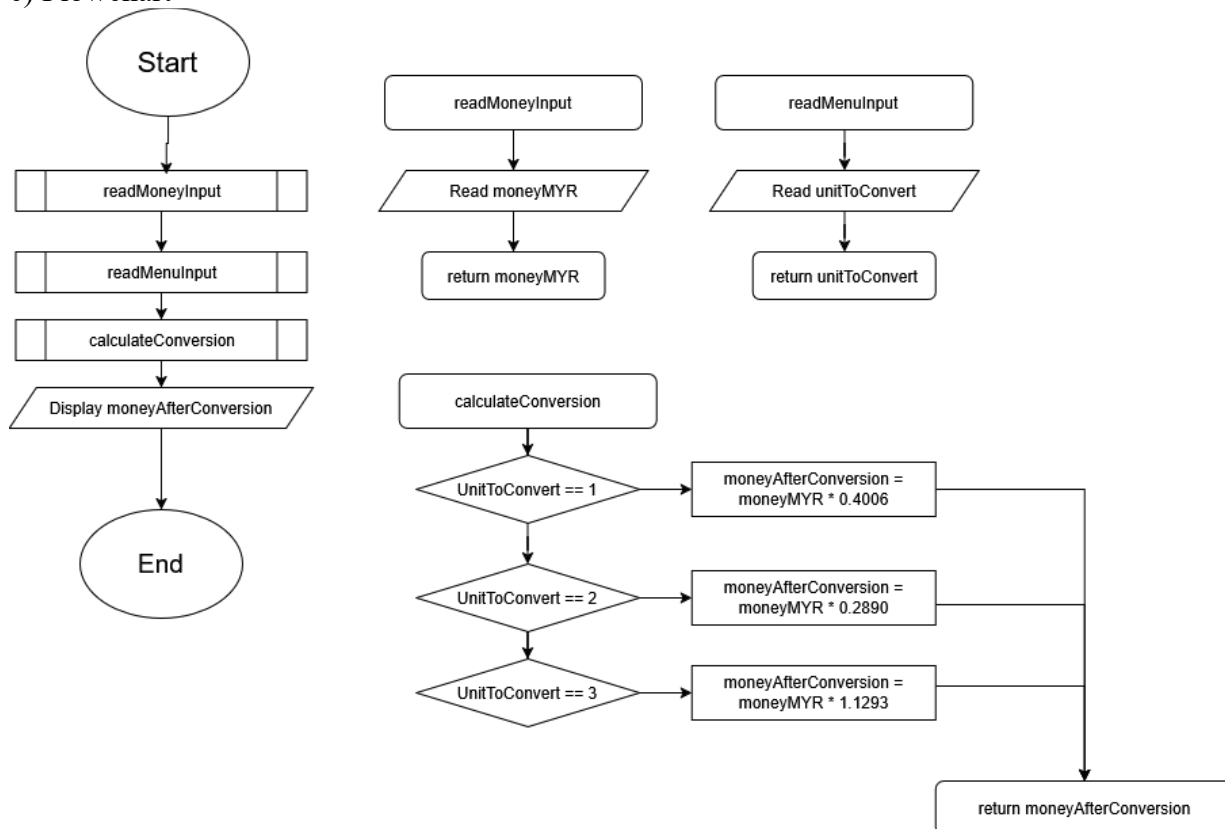
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. MoneyMYR 2. UnitToConvert	1. MoneyAfterConversion
Processing Required	Solution Alternative
Read MoneyMYR, UnitToConvert if(UnitToConvert == 1) moneyAfterConversion = moneyMYR * 0.4006; else if(UnitToConvert == 2) moneyAfterConversion = moneyMYR * 0.2890; else if(UnitToConvert == 3) moneyAfterConversion = moneyMYR * 1.1293; Display moneyAfterConversion	1. Define MoneyMYR as constants. 2. Define MoneyMYR as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. MoneyMYR 2. UnitToConvert	1. Method to read MoneyMYR 2. Method to read UnitToConvert 3. Method to conversion 4. Display MoneyAfterConversion 5. End	1. readMoneyInput 2. readMenuInput 3. calculateConversion	1. MoneyAfterConversion

c) Flowchart



d) Pseudocode

Main Module

```
    Process readMoneyInput
    Process readMenuInput
    Process calculateConversion
    Display moneyAfterConversion
```

End

readMoneyInput Module

```
    Read moneyMYR
    Return moneyMYR
```

Exit

readMenuInput Module

```
    Read unitToConvert
    Return unitToConvert
```

Exit

calculateConversion Module

```
    if(UnitToConvert == 1)
        moneyAfterConversion = moneyMYR * 0.4006
    else if(UnitToConvert == 2)
        moneyAfterConversion = moneyMYR * 0.2890
    else if(UnitToConvert == 3)
        moneyAfterConversion = moneyMYR * 1.1293
```

Exit

e) Java code

```
import java.util.Scanner;
import java.util.InputMismatchException;
// NGO YUK HANG 228991 LAB 7 Q%
public class MoneyConverter{
    public static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args){
        while(true){
            displayMenu();
            int state = acceptMenuInput();
            if(state == 4){
                System.out.println("Thank you for using the program.");
                System.exit(0);
            }
            displayResult(state);
            System.out.println("\nPress enter to continue ... \n");
            scanner.nextLine();
            scanner.nextLine();
        }

        public static void displayMenu(){
            System.out.println("-----");
            System.out.println("Select a number to continue.");
            System.out.println("1. Convert to USD (United State Dollar)");
            System.out.println("2. Convert to GBP (British Pound Sterling)");
            System.out.println("3. Convert to SAR (Saudi Riyal)");
            System.out.println("4. Quit the program");
            System.out.println("-----");
        }

        public static int acceptMenuInput(){
            int input;
            while(true){
                try{
                    input = scanner.nextInt();
                } catch (InputMismatchException e){
                    System.out.println("Error! Please enter number 1,2,3 or 4 only to access the menu.");
                    scanner.nextLine();
                    continue;
                }
                if (!(input == 1 || input == 2 || input == 3 || input == 4)){
                    System.out.println("Error! Please enter number 1,2,3 or 4 only to access the menu.");
                    scanner.nextLine();
                    continue;
                }
                break;
            }
            return input;
        }

        public static double acceptMoneyInput(){
            double input;
            while(true){
                System.out.println("MYR : ");
                try{
                    input = scanner.nextDouble();
                } catch (InputMismatchException e){
                    System.out.println("Error! Please enter valid value.");
                    scanner.nextLine();
                    continue;
                }
                if (input < 0){
                    System.out.println("Error! Please enter valid value.");
                    scanner.nextLine();
                    continue;
                }
                break;
            }
            return input;
        }

        public static void displayResult(int state){
            double moneyMYR = acceptMoneyInput();
            double moneyAfterConversion;
            if (state == 1){
                moneyAfterConversion = moneyMYR * 0.4006;
                System.out.println("MYR " + String.format("%.2f",moneyMYR) + " = USD " + String.format("%.2f",moneyAfterConversion));
            } else if (state == 2){
                moneyAfterConversion = moneyMYR * 0.2890;
                System.out.println("MYR " + String.format("%.2f",moneyMYR) + " = GBP " + String.format("%.2f",moneyAfterConversion));
            } else if (state == 3){
                moneyAfterConversion = moneyMYR * 1.1293;
                System.out.println("MYR " + String.format("%.2f",moneyMYR) + " = SAR " + String.format("%.2f",moneyAfterConversion));
            }
        }
    }
}
```